First Data®

BUYPASS® Platform
ATL105 Specification Updates for TransArmor® Processing

Version: 2024-2

August 09, 2024

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Copyright

© 2024 First Data Corporation

All rights reserved. All information contained herein is confidential and proprietary to First Data Corporation. It shall not be disclosed, duplicated, or used in part or in whole, for any purpose without prior written consent from First Data Corporation. All trademarks, service marks and trade names referenced herein are the property of their respective owners.

Disclaimer

The TransArmor Addendum should be used in conjunction with BUYPASS® Platform ATL105 Message Format Specifications.

Please subscribe to this TransArmor Addendum by checking the subscribe button on the First Data specification website prior to download, so notification will be emailed when new versions are available. Please send an email to certsupport@firstdata.com for additional information.

Current Revision History

This is a new document. It replaces the following document:

- *BUYPASS® Platform ATL105 Specification Updates for TransArmor Processing*
  Version No. 2024-1
  Released on February 09, 2024

The following table lists added, updated, and deleted pages:

Note: Red text and change bars in the page margin indicate updated/added text. Deleted text is indicated only in the following table. Blue text indicates a link to another location in the text.

Changes Made to Ver. No. 2024-1 And Reflected in Ver. No. 2024-2 Released August 09, 2024			
Topic	**2024-1 P. No.**	**2024-2 P. No.**	**Summary of Change**
TA-VE Support EDATA Identifier Value Update	NA	44	Added EDATA identifier value (TA-VE support) '5' in TransArmor - VeriFone Edition (Encryption and Tokenization)—Initial Transaction—Request under Impacted Data Elements

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Prior Revision History

Changes Made to Ver. No. 2023-1 And Reflected in Ver. No. 2024-1 Released February 09, 2024			
Topic	**2023-1 P. No.**	**2024-1 P. No.**	**Summary of Change**
Real Time Account Updater Updates	NA	51	Added section on Real Time Account Updater in data element 2 under Impacted Data Elements
	NA	53	Added section on Real Time Account Updater in data element 12 under Impacted Data Elements

Changes Made to Ver. No. 2020-1 And Reflected in Ver. No. 2023-1 Released August 18, 2023			
Topic	**2020-1 P. No.**	**2023-1 P. No.**	**Summary of Change**
INVALD CAID Updates	NA	30	Added note on restarting the key exchange process under TransArmor® CA Key Exchange Process Flow
	NA	34	Added verbiage on restarting the key exchange process under Certificate Authority Key Rotation

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Table of Contents

1. Introduction 1
1.1 Chapter Conventions 1
1.2 List of Acronyms 3
2. Overview 5
2.1 What is TransArmor®? 5
2.2 Why is TransArmor® Highly Recommended? 5
2.3 TransArmor® Data Protection 6
2.4 TransArmor® Security Solutions 6
3. TransArmor® Encryption and Tokenization 7
3.1 TransArmor® Encryption Methods 7
3.1.1 Supported Data Entry Methods—PKI Encryption and Tokenization 9
3.2 TransArmor® Tokenization 10
3.2.1 TransArmor® Tokens 10
3.2.2 TransArmor® Multi-Pay Tokens 10
3.2.3 Supported Data Entry Methods—Token Only 13
3.3 Hardware/ Software Requirements 13
3.4 Obtaining/Maintaining the Key ID 14
3.5 Supported Card Types 14
3.6 Supported Transaction Types 15
3.7 Message Format Impacts—PKI Encryption and Tokenization 16
3.7.1 Financial Transaction Request Impacts 17
3.7.2 Financial Transaction Response Impacts 17
3.7.3 Key and Key ID Load Request 18
3.7.4 Key and Key ID Load Response 19
3.8 Message Format Impacts—Token Only 20
3.8.1 Financial Transaction Request Impacts 20
3.8.2 Financial Transaction Response Impacts 21
3.9 Receipt Requirements 21
3.10 TransArmor® - VeriFone Edition Processing 21
3.10.1 Registering a terminal for VeriFone Encryption 21
3.10.1.1 Device Registration 22
3.10.1.2 Registration Process 22
3.10.1.3 Device Movement 22
4. TransArmor® Message Formats 23

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

4.1.1 TransArmor® Key and Key ID Load 23
4.1.1.1 Request 23
4.1.1.2 Response 25
5. TransArmor® CA Certificate Overview 27
5.1 Certificate Authority Security Components 27
5.2 TransArmor® CA Key Exchange Process Flow 29
5.3 TransArmor® (with Signed Key) Initialization Process Flow 30
5.3.1.1 Step 1: Request and Response for Signing Key Certificate 31
5.3.1.2 Step 2: Request and Response for Signed Key Certificate 33
6. Impacted Data Segment 35
6.1 TransArmor® Load Data Segment 35
7. Impacted Data Elements 37
8. Appendix A. TransArmor VeriFone Edition: Responses to Administrative Command A-1

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

1. Introduction

This document provides information about the requirements for implementing TransArmor transaction processing involving the BUYPASS® platform (BUYPASS) ATL105 message formats (software version ATL105).

The primary audience of this document includes the merchant or vendor implementing TransArmor processing services.

1.1 Chapter Conventions

This document follows certain typographic conventions in text to help you identify special terms and notes. The following table describes these conventions and provides examples of their usage:

Chapter Conventions

Convention	Description	Example
Copyright Information	The copyright symbol for TransArmor is used only in the headings of the main chapters and the sub chapters.	What is TransArmor®? First Data provides an advanced mode of payment card security service through its TA solution.
TransArmor/TA	In headings and introduction of a section, TransArmor is written in full form, however, in the body of the content, the acronym “TA” is used.	First Data provides an advanced mode of payment card security service through its TransArmor (TA) solution. The TA solution is developed through partnership between First Data, VeriFone Systems, and RSA, the Security Division of EMC.
Subsequent Transactions	This is a collective noun phrase used to denote the transactions that are based on the original authorization request.	The list of subsequent transactions is given below: • Completion • Reversal • Void
Bulleted List	All instances of an unordered series of concepts, items or options are represented in the form of bullets.	Following are the benefits of TA: • Payment card security solution that removes sensitive payment card data from the merchant environment to reduce the threat and liability associated with a data breach • Dual-layered solution combining encryption + tokenization technology to fortify the merchant environment with an end-to-end solution that comprehensively protects payment card data

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Chapter Conventions		
Convention	**Description**	**Example**
Numbered List	All instances of sequence of processes, events or steps are represented in the form of bullets.	1. The POS device sends a TA Key and Key ID Load message requesting for Signing Key 2. The POS device receives the Signing Key and the Signing Key ID in the TA Key and Key ID Load response from First Data

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

1.2 List of Acronyms

In this document, all instances of terms, such as, TransArmor/ Data Encryption Key/ Certificate Authority/ Security packet, etc, used as proper and common nouns in sentences in the chapter's body are represented through acronyms. The table below gives you the list of the acronyms, used across the document.

Acronyms	
Acronym	**Description**
TA	TransArmor
DEK	Data Encryption Key
DDK	Data Decryption Key
SP	Security Packet
CA	Certificate Authority
CA-ID	Certificate Authority Identifier
TA-VE	TransArmor –VeriFone Edition
TRSM	Tamper-Resistant Security Module
SRED	Secure-Reading and Exchange of Data
AES	Advanced Encryption Standard
VAR	Value Added Reseller

Note: Terms present in request/response formats and value sets are not represented through acronyms for better understanding.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

This document contains the following chapters and appendices:

1. **Introduction.** This chapter describes the purpose, usage, target audience, chapter conventions and list of acronyms used in this document. It also provides a list and describes each chapter and appendices present in the document.
2. **Overview.** This chapter provides a general overview on TransArmor.
3. **TransArmor® Encryption and Tokenization.** This chapter describes the TransArmor encryption and tokenization methodologies and supported transactions.
4. **TransArmor® CA Certificate Overview.** This chapter describes the CA Certificate Overview and process flows.
5. **Impacted Data Segment.** This chapter describes the Data Segments which have been impacted by TransArmor transaction processing.
6. **Impacted Data Elements.** This chapter describes the Data Elements which have been impacted by TransArmor transaction processing.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2. Overview

This chapter discusses the following TransArmor processing overview:

- What is TransArmor®?
- Why is TransArmor® Highly Recommended?
- TransArmor® Data Protection
- TransArmor® Security Solutions

2.1 What is TransArmor®?

First Data provides an advanced mode of payment card security service through its TransArmor (TA) solution. The TA solution is developed through partnership between First Data, VeriFone Systems, and RSA, the Security Division of EMC. Following are the benefits of TA:

- Payment card security solution that removes sensitive payment card data from the merchant environment to reduce the threat and liability associated with a data breach
- Dual-layered solution combining encryption + tokenization technology to fortify the merchant environment with an end-to-end solution that comprehensively protects payment card data
- The world's most widely used payment security product - safeguarding 400,000+ merchant outlets and 1 billion transactions+ since its launch in 2010
- Flexible solution that can be implemented on First Data or 3rd Party (VeriFone, Equinox, Ingenico, etc.) devices via hardware (VeriFone only) or software-based encryption and tokenization
- PCI DSS compliant solution - TransArmor exceeds PCI requirements
- Easily integrates with a variety of merchant payment applications via Rapid Connect, to let customers pay anyway and anywhere they choose

2.2 Why is TransArmor® Highly Recommended?

During a transaction processing, the cardholder data flows through multiple entities, technologies, and applications. This long chain of processing units increases high risk of data breach at every step of the transaction flow. Generally, the sensitive cardholder data is at risk of being exposed or stolen at the following two steps in the transaction flow:

- Pre-authorization - In this scenario, a merchant captures the customer’s data to send it to the acquirer/processor. Therefore, at this stage the data is either being sent or is waiting to be sent to the acquirer/processor.
- Post-authorization – In this scenario, the acquirer/processer has already sent back the authorization response to the merchant, and the response data is stored in the merchant environment to be used for analytics and other back-office processes.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2.3 TransArmor® Data Protection

TA security system protects the cardholder data at the following stages in processing:

- In transit: - The sensitive data is protected in transit with advanced encryption options that secure data from the moment of swipe throughout the transaction.
- In use: The sensitive data is removed from the card data environment (CDE) after authorization by replacing the sensitive data with a token or randomly generated number.
- At rest: The non-sensitive tokenized card data is stored safely for back-end business operations and customer analytics. Tokenization of card data reduces the risk of loss of data, minimizes brand damage, and builds customer confidence. TransArmor also reduces the security threat that could result in financial liability and litigation.

2.4 TransArmor® Security Solutions

TransArmor offers two Security Solutions:

- Encryption and Tokenization: This method supports the use of card data encryption and the use of Token. TA enabled terminals encrypt the cardholder data that is present in the initial transaction request. In subsequent transactions with the same card, the token received from First Data is used in place of a card's Primary Account Number (PAN).
- Tokenization Only: The POS device must send the PAN in the clear during initial the transaction to get back a token in the response. In subsequent transactions using the same card, the token received from First Data must be used in place of the card's PAN.

Additionally, there are two Encryption options to choose from and two Tokenization Types to choose from, outlined in the matrices in Chapter 3. TransArmor® Encryption and Tokenization.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

3. TransArmor® Encryption and Tokenization

This chapter discusses the following:

- TransArmor® Encryption Methods
- TransArmor® Tokenization
- TransArmor® CA Certificate Overview

3.1 TransArmor® Encryption Methods

- RSA's Public Key Infrastructure (PKI) encryption provides software-based encryption. This method uses the RSA 2048-bit algorithm in software. PKI can be installed on terminals or PC-based POS devices, letting you add the TransArmor solution with little-to-no investment in new or upgraded hardware
- TransArmor - VeriFone Edition (TA-VE) encryption provides hardware-based, format-preserving encryption. This method uses 128-bit AES symmetric key encryption. TA-VE can be installed on standalone and integrated VeriFone devices. The installation of TA usually requires no software-related modification at the POS application level and no extra steps or training for the retailer

TransArmor Encryption Differentiators		
Method	**TransArmor Public Key Infrastructure (PKI)**	**TransArmor VeriFone Edition**
Technology Partner	RSA	VeriFone
Aliases	• Software-based encryption • RSA encryption • Asymmetric encryption	• Hardware-based encryption • Format-preserving • Symmetric encryption
Summary	Uses two separate keys to protect data: a 'public key' and a 'private key'. Public key is used in the merchant environment to encrypt the cardholder data and the Private key is stored in a Host Security Module (HSM) within First Data's secure environment and is used to decrypt the cardholder data.	Uses VeriFone's existing PCI PTS-approved payment devices, encryption is performed within a Tamper-Resistant Security Module (TRSM) using the terminal operating system's PCI-certified Secure-Reading and Exchange of Data (SRED) module at the point where card data is captured.
How its Applied	• Integrated with any PIN Pad • Installing TA is hardware agnostic which means it can be applied to numerous certified devices • Encrypts Track 1, Track 2, and Keyed	• Integrated with VeriFone integrated PIN Pads ONLY • Cannot serve at POS or gateway level • Encrypts PAN and Discretionary

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

TransArmor Encryption Differentiators		
Method	**TransArmor Public Key Infrastructure (PKI)**	**TransArmor VeriFone Edition**
	Data at POS level	Data in tamper resistant hardware
Key Method	Asymmetric Method- Using two keys (Public & Private), TA infrastructure produces a cipher text of 344 bytes of indecipherable code that protects data by transforming into an illegible series of numbers and letters that is easily distinguishable from cardholder data. Public key can only be decrypted by Private key , both of which are safely stored ONLY at First Data	Symmetric Method - The FPE option relies on symmetric encryption keys, i.e., the same key is used to encrypt and decrypt the cardholder data using Advanced Encryption Standard (AES) that uses unique keys per device
Format	344-bytes of data - long prime numbers called keys, only First Data can unlock the public key with the private key	Encrypts data so output is in same length and characters set as input. This type of encryption is called Format-Preserving Encryption (FPE) First 6-digits and last 4-digits left in the clear to allow BIN and routing functions to be performed correctly. It is important to note that TransArmor (RSA) has BIN routing capabilities as well.
Key Management	One key ID per sales channel. Each sales alliance has a specific key ID; therefore, the vendors will have several key IDs depending on the merchant and their merchant acquirer.	One key ID per device. Keys must be 'injected securely' into device at a secure facility, just like that of PIN Debit. This process ensures that the industry required standard of providing split-knowledge and dual-control is met. Keys can be injected via 'VRK/VeriFone Remote Key'. There is a setup process with VeriFone and the process has minimum O/S requirements. There is no specific timeframe or best practice for key rotation for Unique Keys per Device – merchants are encouraged to follow PCI Key Rotation guidelines.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

3.1.1 Supported Data Entry Methods—PKI Encryption and Tokenization

The ATL105 specification supports various card data entry methods for PKI Encryption and Tokenization.

The new sub-element “EDATA Identifier” that is part of Element No. 2 (Account Number) is used to indicate the type of data being processed. For additional information, please refer to chapter 16, “Data Element Descriptions” in the latest version of ATL105 Message Format Specifications.

PKI Encryption and Tokenization		
Data Entry Method	**EDATA Identifier**	**Description**
Original Transaction Data—Swiped card	1	Encrypted Track1 data
	2	Encrypted Track2 data
Original Transaction Data—Radio Frequency Identification (RFID) Service	1	Encrypted Track1 data
	2	Encrypted Track2 data
Original Transaction Data—Manually keyed **Note**: Expiration Date must be included in Card Discretionary Block Data (Element No. 12).	3	Encrypted PAN
Follow-on Transaction Data **Note**: Expiration Date must be included in Card Discretionary Block Data (Element No. 12).	0	Host-assigned Token

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

3.2 TransArmor® Tokenization

First Data has two types of tokens available, TransArmor Multi-Pay Tokens and TransArmor Tokens, each with different capabilities, however, all Tokens share the below common characteristics:

- Tokens match the format of the initiating PAN. For example: If the PAN is of 19-digits, the token is also of 19-digits length
- Tokens are unique for major card brands (American Express, Discover, MasterCard, or Visa)
- Tokens do not pass MOD-10 or Luhn checks
- Tokens share the same last 4-digits as those of the corresponding PAN
- Tokens do not expire. The same token follows the card through all the subsequent transactions
- Tokens are card-based. A merchant will always receive a unique token for a specific PAN
- Token Values are non-reversible and not mathematically derived from the initiating PAN
- Tokens can be shared universally or are unique to a merchant (MultiPay Token)

3.2.1 TransArmor® Tokens

TransArmor Tokens are generated at the RSA Safe Proxy within FD's PCI secure environment and returned with the transaction's authorization response to the merchant. Tokens cannot be used to initiate financial transactions. The Token is returned as part of the authorization process and the merchant is able to make adjustments to the transaction until the end of day settlement is sent to First Data.

3.2.2 TransArmor® Multi-Pay Tokens

TransArmor Multi-Pay Tokens address the primary security risk and PCI burden inherent in a Card-Not-Present (CNP) environment—the need to store customer's preferred payment information for subsequent transactions (using the same card)—making them an ideal solution for e-Commerce merchants and service providers with recurring invoices. Multi-Pay Tokens have the same characteristics as TransArmor Token, plus the following additional features:

- Unique to a particular merchant
- Replace cards' PANs in all subsequent transactions (card present and card not present) using the same card – valuable for both POS and e-Commerce/ CNP environments
- Stored in lieu of the PAN to perform transactions for new or recurring payments
- Used for refunds and credits also
- Enables merchants to track buying patterns (buying pattern is used to analyze the sales trends) and launch marketing/loyalty programs while within PCI Compliance
- Supports any business that needs to initiate a financial transaction without card being present

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

TransArmor Tokenization Differentiators		
Feature	**TransArmor Token**	**Multi-Pay Token**
Technology Partner	RSA (SafeProxy)	RSA (SafeProxy)
Aliases	• Single Use • Former Name: F-Token • Token Type must be “0001”	• Multi–Use • Former Name: R-Token • Token Type is not “0001”
Summary	Process of replacing a PAN with a “token”, a non-sensitive surrogate. The token uniquely represents a customer’s account information, removing the need for merchants to store PAN’s in their card data environment (CDE) Tokenization mitigates security risks associated with storing sensitive card payment data on merchant system	Same as TransArmor Token
Token Assignment	Shared across ALL TransArmor merchants	Unique for each merchant
Format	Format-preserving, meaning they match the format and share last four digits of the initiating PAN. If a card is 16 digits in length, Token would be a random set of 12 numbers plus the last 4 digits left in the clear. Tokens are generated and distributed specific to Card, so a merchant will always get the same token back for a specific PAN. Generated to not overlap major brand (Visa, MC, AMEX, Discover) BIN ranges (first digit is 0-2 or 7-9) Do not expire – follows the PAN through the card lifecycle	Same as TransArmor Token
Transactions with Token	• Cannot perform any financial transactions with the token itself • Voids (within same day)	• Supports recurring payments from customer, the merchant can submit a Multi-Pay Token to initiate a sale on a regular basis for recurring transactions without contacting the customer to reconfirm

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

TransArmor Tokenization Differentiators		
Feature	**TransArmor Token**	**Multi-Pay Token**
		card data • Perform any financial transaction (including credits, returns, and sales) with Token itself Can get Token via a non-financial transaction to place in customer profile for future use • Can only be monetized by the merchant

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

3.2.3 Supported Data Entry Methods—Token Only

The ATL105 specification supports various card data entry methods for Token Only transactions.

The new sub-element “EDATA Identifier” that is part of Element No. 2 (Account Number) is used to indicate the type of data being processed. For additional information, please refer to chapter 16, “Data Element Descriptions” in the latest version of ATL105 Message Format Specifications.

Token Only		
Data Entry Method	**EDATA Identifier**	**Description**
Original Transaction Data—Swiped card	4	Unencrypted Track1 or Track2 data
Original Transaction Data—Radio Frequency Identification (RFID) Service	4	Unencrypted Track1 or Track2 data
Original Transaction Data—Manually keyed **Note**: Expiration Date must be included in Card Discretionary Block Data (Element No. 12).	4	Unencrypted PAN
Follow-on Transaction Data **Note**: Expiration Date must be included in Card Discretionary Block Data (Element No. 12).	0	Host-assigned Token

3.3 Hardware/ Software Requirements

The following information applies to both processing methods.

In order to use TransArmor, a merchant must have the required POS device and the software release—either PKI Encryption and Tokenization or Token Only—that support this feature.

Note: Contact your BUYPASS representative for a current listing of equipment options.

TransArmor is an optional software feature. Vendors developing code to support it must have the capability to enable/disable it.

Merchant participation in TransArmor must be approved by First Data Corporation followed by the necessary certification process. Only then can it be enabled/disabled by the vendor. Please contact your BUYPASS representative for additional information.

Note: Only supported cards can be used for TransArmor. For a list of supported cards, please see section 3.5, “Supported Card Types.”

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

3.4 Obtaining/Maintaining the Key ID

Note: Obtaining/Maintaining the Key ID can take place only for PKI Encryption and Tokenization.

The following information applies only to PKI Encryption and Tokenization.

Before successful TransArmor processing can occur, the device must have the appropriate Key ID. The device downloads the Key ID following hardware and/or software installation.

TransArmor processing takes place after the Key ID has been downloaded using the TransArmor Key Load process.

The Key ID is maintained under the following circumstances:

- Faulty equipment replacement
- Regular Key ID update schedules
- Nonscheduled Key ID updates as a result of an unsuccessful TransArmor transaction that indicates that a new Key ID and Key are needed

3.5 Supported Card Types

The following card types are supported for both processing methods:

- Credit
- Debit
- Electronic Benefits Transfer (EBT)
  - ◊ Food stamp benefits
  - ◊ Cash benefits
  - ◊ eWIC

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

3.6 Supported Transaction Types

These specifications support Financial Transaction Requests and Responses for the following transaction types (These transactions have been arranged in groups based on similar transaction flow) for both processing methods:

Note: Please refer to section “Message Format Impacts—PKI Encryption and Tokenization”

and section 3.8, “Message Format Impacts—Token Only,” for specific message format information.

Supported Transaction Types—Original Transactions	
Transaction Type Code	**Transaction Type**
0	POS Purchase
3	POS Authorization Only
5	CAT Authorization Only
7	Merchandise Return **Note:** Returns can use the Token or the card’s account information.
B	Mail/Phone Authorization Only

Supported Transaction Types—Follow-on Transactions	
Transaction Type Code	**Transaction Type**
0	POS Capture (Completion)
4	CAT Capture (Completion)
6	Mail/Phone Purchase
8	Purchase Reversal
C	Mail/Phone Reversal
S	Cancellation
U	Void of Merchandise Return

Supported Transaction Types—Time-out Reversal Transaction	
Transaction Type Code	**Transaction Type**
Z	Time-out Reversal

Supported Transaction Types—Token Registration Transaction	
Transaction Type Code	**Transaction Type**
T	GET-TOKEN-FOR-PAN

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

3.7 Message Format Impacts—PKI Encryption and Tokenization

The following existing messages are impacted by TransArmor PKI Encryption and Tokenization:

- TransArmor PKI Encryption and Tokenization Financial Transaction Request
- TransArmor PKI Encryption and Tokenization Financial Transaction Response

The following new messages have been added for TransArmor PKI Encryption and Tokenization: These messages also support CA Signed Key and Signing Key ID loads.

- TransArmor Key and Key ID Load Request
- TransArmor Key and Key ID Load Response

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

3.7.1 Financial Transaction Request Impacts

The following table summarizes the data sections, segment, and elements that must be present with the appropriate TransArmor PKI Encryption and Tokenization information for this message:

Notes: For detailed message layout information, please refer to section 14.1.1, “Request” In the latest version of ATL105 Message Format Specifications.

For detailed data segment information, please refer to section 15.1, “Standard Message Data Segment” in the latest version of ATL105 Message Format Specifications.

For detailed data element information, please refer to section 16.2, “Data Elements in Data Element Number Order” in the latest version of ATL105 Message Format Specifications.

TransArmor PKI Encryption and Tokenization **Financial Transaction Request—Impacts** **Required Data Sections, Segment, and Elements**	
Data Section No. 1	**Data Section No. 2** **Data Segment 100 (Standard Message Data Segment)**
• Message Format Version Identifier (Element No. 55)	• Account Number (Element No. 2) • Card Discretionary Block Data[1] (Element No. 12)

[1]For an initial manually keyed transaction and all follow-on transactions, the expiration date must be included in Card Discretionary Block Data (Element No. 12).

3.7.2 Financial Transaction Response Impacts

The following table lists the data section and elements that must be present with the appropriate TransArmor PKI Encryption and Tokenization information for this existing message:

Notes: For detailed message layout information, please refer to section 14.1.2 “Response” in the latest version of ATL105 Message Format Specifications.

For detailed data element information, please refer to section 16.2, “Data Elements in Data Element Number Order” in the latest version of ATL105 Message Format Specifications.

TransArmor PKI Encryption and Tokenization **Financial Transaction Response—Impacts** **Required Data Section and Elements** **Data Section No. 1**
• Download Indicator (Element No. 30) • Account Number (Element No. 2) • Decline Code (Element No. 26) • Terminal Display /Printer Message (Element No. 101)

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

3.7.3 Key and Key ID Load Request

The Key and Key ID Load Request is a new message for TransArmor PKI Encryption and Tokenization. It is used to request a new TransArmor Key and Key ID.

Note: Key and Key ID or CA Signed Key and Signing Key ID loads must be supported when using TransArmor PKI Encryption and Tokenization method.

The following table summarizes the data sections, segment, and elements that must be present with the appropriate TransArmor PKI Encryption and Tokenization information for this message:

Notes: For detailed message layout information, please refer to section 4.1.1.1, “Request” in this document.

For detailed data segment information, please refer to section 6.1, “TransArmor Load Data Segment” in this document.

For detailed data element information, please refer to section 16.2, “Data Elements in Data Element Number Order” in the latest version of ATL105 Message Format Specifications.

TransArmor PKI Encryption and Tokenization Key and Key ID Load Request Required Data Sections, Segment, and Elements	
Data Section No. 1	**Data Section No. 2 Data Segment 116 (TransArmor Key Load Data Segment)**
• Message Format Version Identifier (Element No. 55) • Number of Segments (Element No. 63)	• Segment Type (Element No. 85) • Segment Length (Element No. 84) • Sequence Number (Element No. 86) • Terminal Identifier (Element No. 102) • Load Type (Element No. 48) • Hardware Version (Element No. 43) • Software Version (Element No. 96) • Firmware Version (Element No. 39)

**This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.**

3.7.4 Key and Key ID Load Response

This message is used to return a new TransArmor Key and Key ID.

Note: Key and Key ID or CA Signed Key and Signing Key ID loads must be supported when using TransArmor PKI Encryption and Tokenization method.

The following table summarizes the data elements that must be present with the appropriate TransArmor PKI Encryption and Tokenization information for this message:

Notes: For detailed message layout information, please refer to section 4.1.1.2, “Response” in this document.

For detailed data element information, please refer to section 16.2, “Data Elements in Data Element Number Order” in the latest version of ATL105 Message Format Specifications.

TransArmor PKI Encryption and Tokenization **Key and Key ID Load Response** **Required Data Elements**
• Data Type Indicator (Element No. 24) • Sequence Number (Element No. 86) • Response Code (Element No. 83) • Key ID (Element No. 155) • Key Data Length (Element No. 156) • Key Data (Element No. 157)

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

3.8 Message Format Impacts—Token Only

The following is the new message specific to TransArmor Multi-Pay Token transactions:

- TransArmor Token Registration (GET-TOKEN-FOR-PAN)

The following existing messages are impacted by TransArmor Token Only:

- TransArmor Token Only Financial Transaction Request
- TransArmor Token Only Financial Transaction Response

3.8.1 Financial Transaction Request Impacts

The following table summarizes the data sections, segment, and elements that must be present with the appropriate TransArmor Token Only information for this message:

Notes: For detailed message layout information, please refer to section 14.1.1, “Request” in the latest version of ATL105 Message Format Specifications.

For detailed data segment information, please refer to section 15.1, “Standard Message Data Segment” in the latest version of ATL105 Message Format Specifications.

For detailed data element information, please refer to section 16.2, “Data Elements in Data Element Number Order” in the latest version of ATL105 Message Format Specifications.

TransArmor Token Only Financial Transaction Request—Impacts Required Data Sections, Segment, and Elements	
Data Section No. 1	**Data Section No. 2 Data Segment 100 (Standard Message Data Segment)**
• Message Format Version Identifier (Element No. 55)	• Account Number (Element No. 2) • Card Discretionary Block Data[1] (Element No. 12)

[1]For an initial manually keyed transaction and all follow-on transactions, the expiration date must be included in Card Discretionary Block Data (Element No. 12).

**This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.**

3.8.2 Financial Transaction Response Impacts

The following table summarizes the data section and elements that must be present with the appropriate TransArmor Token Only information for this existing message:

Notes: For detailed message layout information, please refer to section 14.1.2, “Response” in the latest version of ATL105 Message Format Specifications.

For detailed data element information, please refer to section 16.2, “Data Elements in Data Element Number Order” in the latest version of ATL105 Message Format Specifications.

TransArmor Token Only **Financial Transaction Response—Impacts** **Data Section No. 1** **Required Data Elements**
• Account Number (Element No. 2) • Terminal Display /Printer Message (Element No. 101)

3.9 Receipt Requirements

In addition to the receipt requirements described in section 10.1.7, “Receipt Requirements,” of the latest version of ATL105 Message Format Specifications, the following TransArmor requirement is necessary:

- The masked number printed on all receipts is the masked token. The last four digits of the token are the same as the last four digits of the card’s PAN.

3.10 TransArmor® - VeriFone Edition Processing

First Data and VeriFone Systems, Inc. are working together to offer a VeriFone edition of the First Data TransArmor solution to US multi-lane and petroleum merchants. This will enable businesses using VeriFone Mx devices and other devices to take advantage of a complete security solution combining VeriFone’s encryption along with tokenization technology from RSA, the Security Division of EMC Corporation.

Note: Contact your BUYPASS representative for a current listing of device options.

Please refer to the following new information about TransArmor - VeriFone edition processing (TA-VE) in the latest version of ATL105 Message Format Specifications:

- Section 15.11, “Additional Information Data Segment”
- Element No. 2 (Account Number)
- Element No. 116 (Additional Information Indicator)
- Element No. 118 (Additional Information)
- Appendix K, “Additional Information Data Layouts”

3.10.1 Registering a Terminal for VeriFone Encryption

In order to use TransArmor VeriFone encryption, the device must first be registered by initiating a Registration event which activates the VeriFone encryption on the terminal and registers the device with the encryption/decryption server.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

3.10.1.1 Device Registration

Device Registration refers to the process of using the "RegiStart" Card or RegiStart administrative function coded for the POS. This enables encryption on the device and registers the device with the encryption/decryption server. The RegiStart Card (or function) is encoded with Track 3 data unique to the Merchant's boarding information and the device configuration file. The encrypted PAN of the RegiStart card/function is 15 digits (within ISO standards). Additionally, the BIN is 515111.

3.10.1.2 Registration Process

- Ensure the device is connected to the point of sale and powered on.
- Initiate a $1.00 MasterCard Sale transaction with the payment application and swipe the RegiStart Card or initiate the RegiStart administrative function coded on the POS. The transaction should be sent using the same TAP information as will be used when sending normal transactions.
- The MasterCard Sale transaction is format ted and message is preserved as Track 1 and Track 2 data.
- Buypass sends the transaction to the decryption server.
- The Terminal ID is registered with the decryption server during the registration process.
- The MasterCard Sale transaction Response Code is sent back to the merchant.
- The device is now registered and is in an encrypting state ready to encrypt transactions.

3.10.1.3 Device Movement

When a device moves from one lane to another, whether previously encrypting or not, or is brand new out of the box, it must be re-registered. Re-register the device by performing the steps listed above in the Registration Process. For more information, please contact your account representative.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

4. TransArmor® Message Formats

4.1.1 TransArmor® Key and Key ID Load

This section contains request and response message formats for a TA Key and Key ID Load.

Note: Key and Key ID or CA Signed Key and Signing Key loads must be supported when using TA PKI Encryption and Tokenization method.

4.1.1.1 Request

The following table contains format information about the TA Key and Key ID Load Request:

Note: Field Separators are present between Field No. 2 and Data Section No. 2; Field Nos. 3 and 4; 4 and 5; and 5 and 6. A Field Separator follows Field No. 10. There are no Field Separators between Field Nos. 6 and 7; 7 and 8, 8 and 9; and 9 and 10; and 10 and 11 when present.

Message Format **TransArmor Key and Key ID Load Request**						
Data Section No. 1						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	55	Message Format Version Identifier	1	6	R	Indicates the message format version. **Fixed value**: ATL105 **Source**: Device
2	63	Number of Segments	7	2	R	Indicates the number of segments in the message. **Fixed value**: 01 **Source**: Device
<FS>						
Data Section No. 2 **Note:** This request does not contain the Standard Message Data Segment (Data Segment No. 100).						
Data Segment No. 116 (TransArmor Key Load Data Segment) **Maximum Segment Length:** 50 bytes						
3	85	Segment Type	9	3	R	Identifies the segment. **Fixed value**: 116 **Source**: Device
<FS>						
4	84	Segment Length	12	3	R	Describes the segment's length. **Variable Length**: 01—66 **Source**: Device
<FS>						
5	86	Sequence Number	15	6	R	Identifies the merchant-assigned Sequence Number for the transaction. **Source**: Device

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Message Format TransArmor Key and Key ID Load Request						
Data Section No. 1						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
<FS>						
6	102	Terminal Identifier	21	13	R	Identifies the transaction device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. For TransArmor, the first two characters (Device Type) of this element's value must be "++" regardless of the actual device type. **Source**: Device
7	48	Load Type	34	1	R	Describes the load type. **Fixed value**: K (Indicates that PKI/PKI Signed Key information is requested.) S (CA Signing Key information is requested.) **Source**: Device
8	43	Hardware Version	35	4	R	Identifies the version number of the device's hardware level. **Source**: Device
9	96	Software Version	39	8	R	Identifies the version number of the device's software application. **Source**: Device
10	39	Firmware Version	47	8	R	Identifies the version number of the device's EPROM level. **Source**: Device
<FS>						
11	155	Key ID	55	11	C	Identifies the CA Key ID associated with the Signed Key being requested. This is a two byte ID followed by 9 spaces. This field is present only when a signed PKI key is being requested. **Source**: Device
<FS>						

[1]Required/Optional/Conditional

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

4.1.1.2 Response

The following table contains format information about the TA Key and Key ID Load Response:

Note: There are no Field Separators in this message.

Message Format TransArmor Key and Key ID Load Response						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	24	Data Type Indicator	1	1	R	Describes the returned data type. **Fixed value**: & (Indicates a new Key and Key ID are being returned.) **Source**: Host
2	86	Sequence Number	2	6	R	Must be echoed from the TransArmor Key and Key ID Load Request. **Source**: Request
3	83	Response Code	8	1	R	Indicates whether a TransArmor transaction has been approved/rejected. **Valid Values**: K (Indicates an approved TransArmor transaction.) S (Indicates a declined TransArmor transaction) P (Indicates a successful Signing/Signed Key Load request) L (Indicates either a rejected TransArmor CA-ID transaction or a merchant who does not participate in TransArmor.) **Source**: Host
4	155	Key ID	9	11	C	Identifies the CA Key ID or PKI Key ID associated with the Signing Key/Signed in Data Element No. 157 (Key Data). For Signing Key, this element is of 2 bytes, followed by 9 spaces. For Signed Key, this element is of 11 bytes. **Source**: Host
5	156	Key Data Length	20	3	C	Identifies the length of the following field (Key Data). **Valid Values**: 000—999 **Source**: Host

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Message Format TransArmor Key and Key ID Load Response						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
6	157	Key Data	23	999	R	Identifies the CA Signing Key associated with Key ID (Data Element No. 155) or the Error Message. For an approved TransArmor transaction, the maximum length is 999 bytes. For a declined TransArmor transaction, a message is returned in this field and is displayed at the device. **Valid Value**: Any valid key **Source**: Host
< ~ >						

[1]Required/Optional/Conditional

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

5. TransArmor® CA Certificate Overview

First Data has implemented the TransArmor (TA) Certificate Authority (CA) solution to handle key exchange for vendor POS terminals and VARs. Certificate Authority solution will enable First Data to validate the source of the public keys.

The standard approach to key delivery is to use a signed X.509 Certificate. The salient features of PKI are listed below:

- First Data sends the Public Data Encryption Keys (DEK) to the merchant through X.509 signed security certificates. The keys are embedded in those certificates.
- The POS terminal is embedded with a CA Certificate chain, so that it can verify and validate the signed DEK.
- Upon initialization, the POS terminals download these signed certificates.

Note: The CA Certificate chain includes the following:

- Signing Key Certificate
- Signed Key Certificate

5.1 Certificate Authority Security Components

The following security key components are required at the terminal in order to support the enhanced TA secure key exchange process.

- Root CA Certificate
- Boot Verification Key (public)
- Signing Key (private)
- Working Verification Key (public)
- Signing Key ID
- Signed Key
- Data Encryption Key (public)
- Data Encryption Key ID
- Data Decryption Key (private)

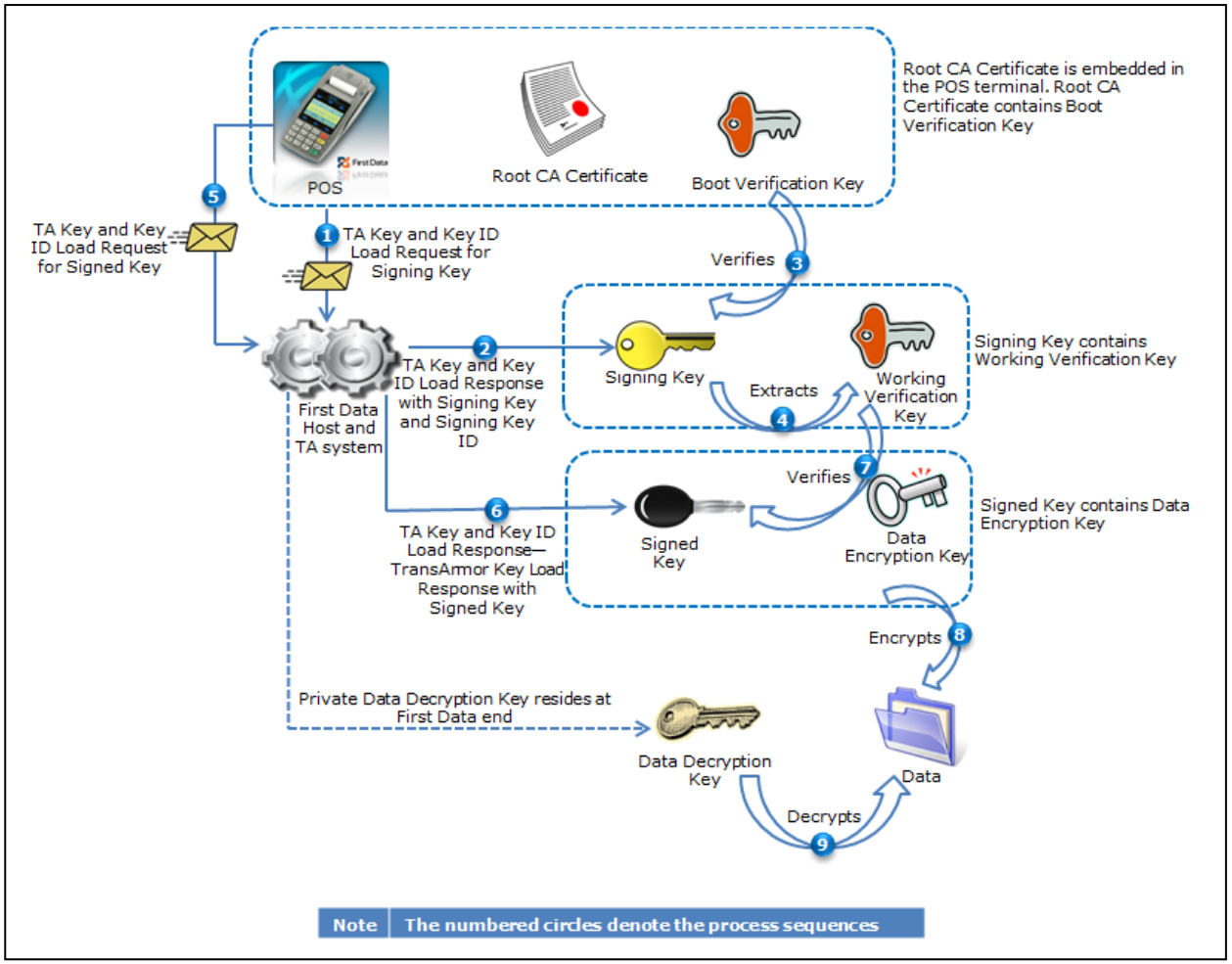
This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Component	Description and Usage
Root CA Certificate	X.509 certificate that contains the Boot Verification Key. The duration of validity is defined in the Certificate.
Boot Verification Key (public)	Public key derived from the Root CA Certificate embedded in the POS device. It is created by First Data Root CA. It used to validate the signature on the Signing Key, in order to unlock the Working Signing Key.
Signing Key (private)	Private key associated with the Working Verification Key (public key). This key is used to sign the Signed Key. It is created by First Data Intermediate Certificate Authority.
Working Verification Key (public)	Derived at POS device from the Signing Key. It is used to validate the signature on the Signed Key which unlocks the Data Encryption Key (DEK).
Signing Key ID	Key ID associated with the Working Signing Key/Working Verification Key private/public key pair. It is sent in request for the Data Encryption Key (DEK) to indicate that the host should send the DEK in an X.509 certificate rather than unsigned.
Signed Key	Key that contains the Data Encryption Key (public). It is signed using the Working Signing Key (private).It is validated using the Working Verification Key (public). The duration of validity is defined in the Certificate. It is downloaded during TransArmor configuration along with the Data Encryption Key ID (DE_ID).
Data Encryption Key (public)	Public key associated with private Data Decryption Key. It is derived at the POS device from the Signed Key. It is used to encrypt the cardholder sensitive data at the POS device.
Data Encryption Key ID	Key ID associated with the Data Decryption Key/Data Encryption Key private/public key pair. It is sent with encrypted data to indicate the Data Decryption Key that must be used to decrypt the data.
Data Decryption Key (private)	Private keys are held by First Data and associated with public Data Encryption Key. It is used to decrypt the cardholder sensitive data at the First Data host.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

5.2 TransArmor® CA Key Exchange Process Flow

The following diagram represents the TA CA security keys exchange process.



1) The POS device sends a TA Key and Key ID Load message requesting for Signing Key.
2) The POS device receives the Signing Key and the Signing Key ID in the TA Key and Key ID Load response from First Data.
3) The POS device uses the Boot Verification Key to validate the Signing Key.
4) After the Signing Key is validated, the Boot Verification Key extracts the Working Verification Key from the Signing Key.
5) The POS device sends a TA Key and Key ID Load message request for Signed Key using the Signing Key ID.
6) The POS device receives the Signed Key in the TA Key and Key ID Load response from First Data.
7) The POS device uses the Working Verification Key to validate the Signed Key.
8) After the Signed Key is validated, the Working Verification Key extracts the DEK.
9) The DEK is used to encrypt the sensitive cardholder data in the TA authorization request.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Note:

- The Root CA Certificate is embedded in the POS terminal.
- The POS terminal extracts the Boot Verification Key from the Root CA Certificate.
- The Root CA is securely managed by the encrypting device and is delivered outside of the message specification.
- If you receive INVALD CAID in the response received in step 2 or 6, restart the key exchange from step 1.

5.3 TransArmor® (with Signed Key) Initialization Process Flow

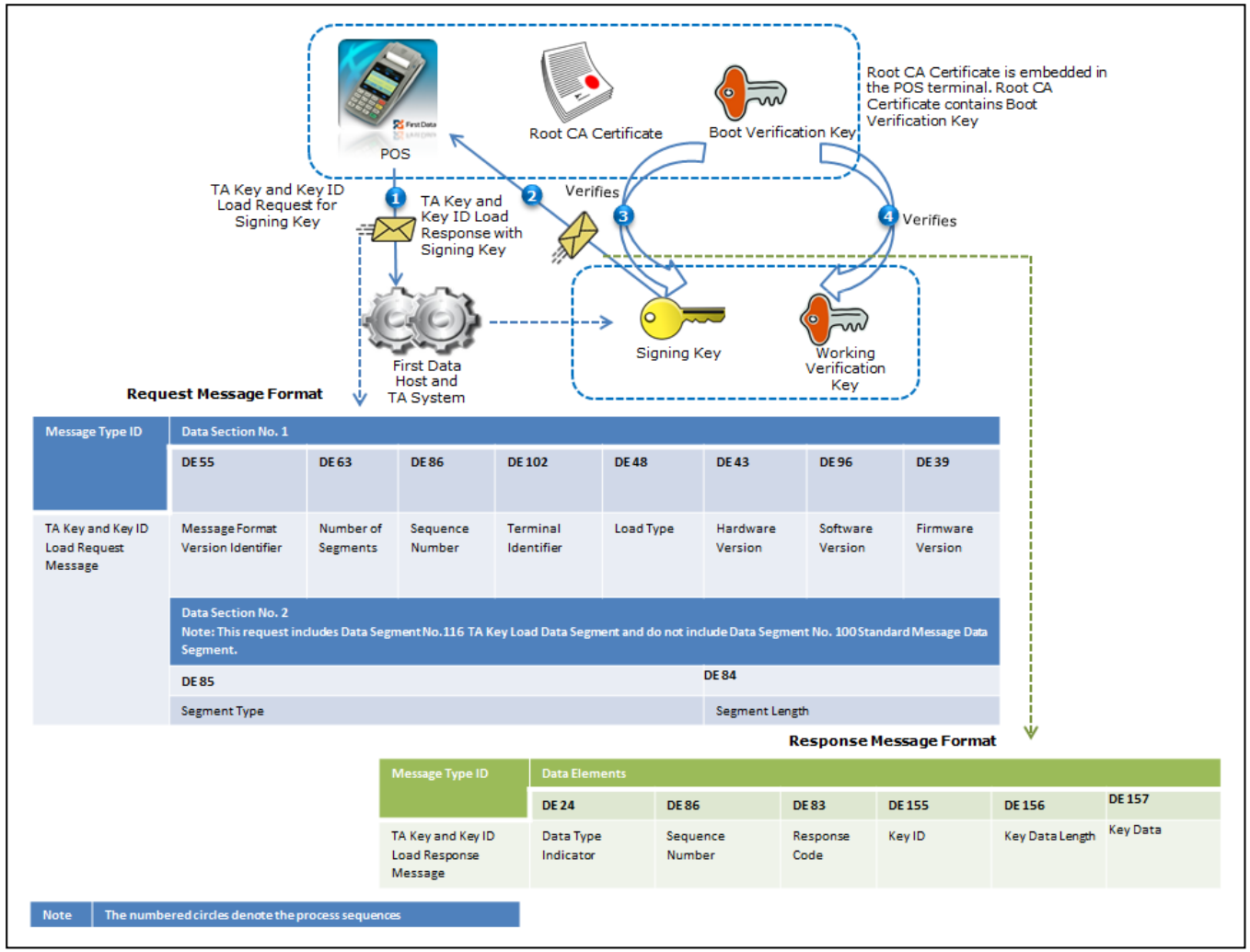
If the terminal vendor has embedded root key and is enabled for Certificate Authority, then the initialization process is as follows below:

Note:

- The Root CA Certificate is embedded in the application download package
- The Root CA Certificate contains the public Boot Verification Key

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

5.3.1.1 Step 1: Request and Response for Signing Key Certificate



Request Message Format

Message Type ID	Data Section No. 1							
	DE 55	DE 63	DE 86	DE 102	DE 48	DE 43	DE 96	DE 39
TA Key and Key ID Load Request Message	Message Format Version Identifier	Number of Segments	Sequence Number	Terminal Identifier	Load Type	Hardware Version	Software Version	Firmware Version
	Data Section No. 2 Note: This request includes Data Segment No.116 TA Key Load Data Segment and do not include Data Segment No. 100 Standard Message Data Segment.							
	DE 85					DE 84		
	Segment Type					Segment Length		

Response Message Format

Message Type ID	Data Elements					
	DE 24	DE 86	DE 83	DE 155	DE 156	DE 157
TA Key and Key ID Load Response Message	Data Type Indicator	Sequence Number	Response Code	Key ID	Key Data Length	Key Data

Note	The numbered circles denote the process sequences

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

1. The POS device sends a TA Key and Key ID Load message requesting for Signing Key.

Note: In the TA Key and Key ID Load Request Message:

- Data Element 48 (Load Type) is set to “S” (CA Signing Key information is requested) in order to request for the Signing Key

2. The POS device receives the Signing Key in the TA Key and Key ID Load response from First Data.

Note: In the TA Key and Key ID Load Response Message:

- Data Element 155 (Key ID) contains the Signing Key
- Data Element 157 (Key Data) contains the Signing Key ID

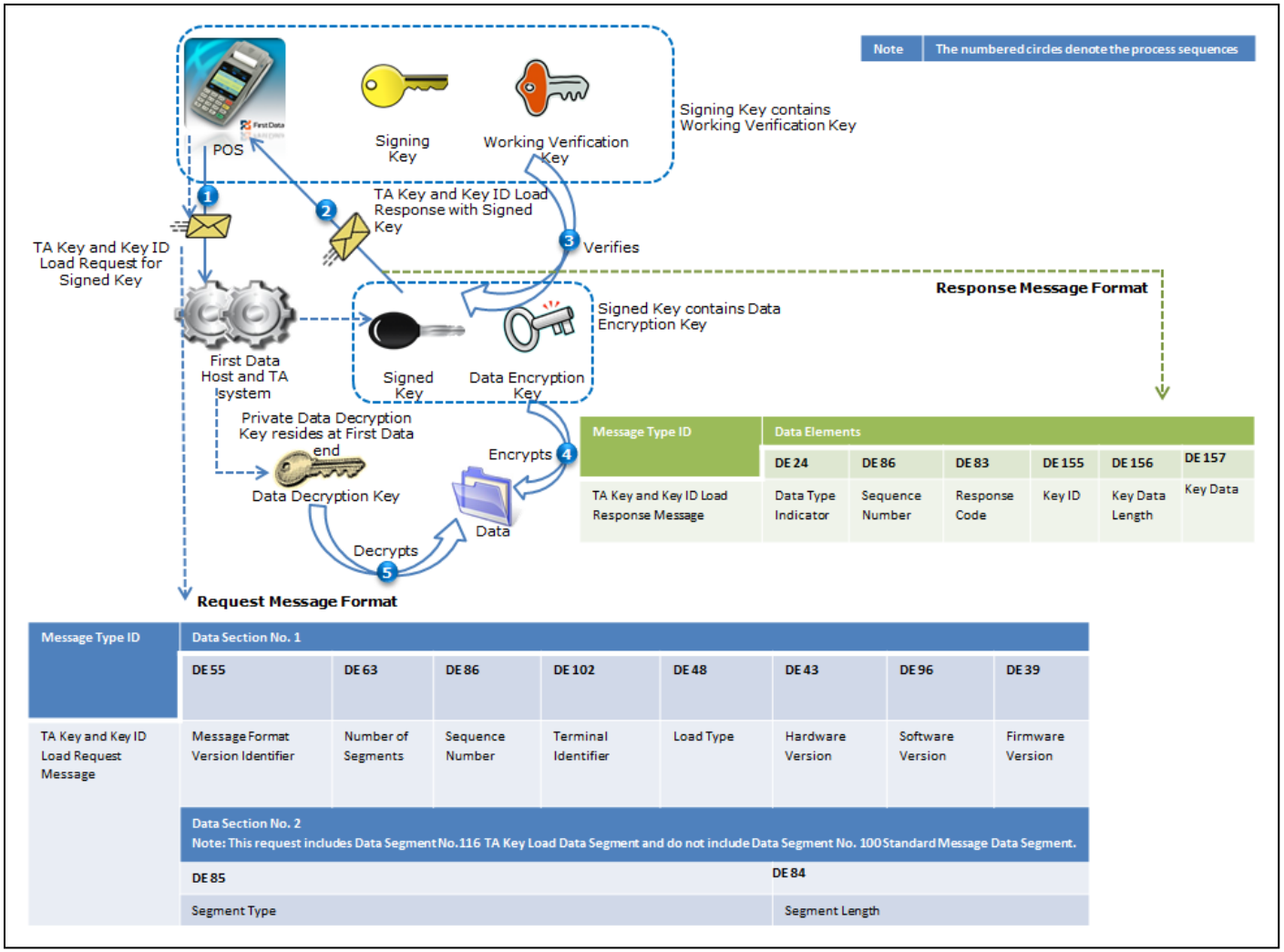
3. The POS device uses the Boot Verification Key to validate the Signing Key.

4. After the Signing Key is validated, the Boot Verification Key validates the Working Verification Key.

Note: The Signing Key contains the Working Verification Key.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

5.3.1.2 Step 2: Request and Response for Signed Key Certificate



Message Type ID	Data Elements					
	DE 24	DE 86	DE 83	DE 155	DE 156	DE 157
TA Key and Key ID Load Response Message	Data Type Indicator	Sequence Number	Response Code	Key ID	Key Data Length	Key Data

Message Type ID	Data Section No. 1							
	DE 55	DE 63	DE 86	DE 102	DE 48	DE 43	DE 96	DE 39
TA Key and Key ID Load Request Message	Message Format Version Identifier	Number of Segments	Sequence Number	Terminal Identifier	Load Type	Hardware Version	Software Version	Firmware Version

Data Section No. 2 Note: This request includes Data Segment No.116 TA Key Load Data Segment and do not include Data Segment No. 100 Standard Message Data Segment.	
DE 85	DE 84
Segment Type	Segment Length

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

1. The POS device sends a TA Key and Key ID Load message requesting for Signed Key Certificate.

Note: In the TA Key and Key ID Load Request Message:

- Data Element 48 (Load Type) is set to “K” (Indicates that PKI/PKI Signed Key information is requested) in order to request for the Signed Key

2. The POS device receives the Signed Key in the TA Key and Key ID Load response from First Data.

Note: In the TA Key and Key ID Load Response Message:

- Data Element 155 (Key ID) contains the Data Encryption Key (DEK) ID
- Data Element 157 (Key Data) contains the Signed Key

3. The POS device uses the Working Verification Key to validate the Signed Key.

4. After the Signed Key is validated, the Working Verification Key validates the DEK.
5. The DEK is used to encrypt the sensitive cardholder data in the TA authorization request.
6. First Data uses the Data Decryption Key (DDK) ID to decrypt the encryption block.

Certificate Authority Key Rotation

- First Data notifies the terminals that they are required to update DEK; the notification comes through value ‘8’ of Download Indicator (Data Element 30) and is received by the terminal
- If the Signing Key ID sent in a request does not match the one that First Data expects, the terminals must download new encryption key
- The CA Certificates and the DEK have an expiration date after which they are no longer accepted by First Data and the transactions would be declined
- If you receive INVALD CAID in the response to a TA Key and Key ID Load Request Message, you must restart the Key Exchange process from Step 1.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

6. Impacted Data Segment

6.1 TransArmor® Load Data Segment

This section describes Data Segment No. 116, TransArmor® Load Data Segment.

It is only sent on TransArmor (TA) Load Request messages to request a new TA Key and Key ID. It contains data elements that are unique to this request. This data segment is also used for CA Signing Key requests.

It always appears at the end of a TA Load Request.

It has a maximum length of 50 alphanumeric characters (01–50/a-z/A–Z).

The device originates this transaction.

No other data segments are contained in the TA Load Request.

The following table illustrates the layout of the TA Load Data Segment:

Note: There is a Field Separator between Field Nos. 1 and 2; 2 and 3; 3 and 4; and following Field No. 8.

TransArmor Load Data Segment Format Data Segment No. 116					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 116 Source: Device
<FS>					
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
<FS>					
3	86	Sequence Number	6	R	Identifies the merchant-assigned Sequence Number for this transaction. Source: Device
<FS>					
4	102	Terminal Identifier	13	R	Identifies the transaction device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. For TransArmor, the first two characters (Device Type) of this element's value must be "++" regardless of the actual device type. Source: Device
5	48	Load Type	1	R	Indicates that Key information is requested. Fixed value: K and S Source: Device

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

TransArmor Load Data Segment Format Data Segment No. 116					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
6	43	Hardware Version	4	R	Identifies the version number of the device's hardware level. Source: Device
7	96	Software Version	8	R	Identifies the version number of the device's software application. Source: Device
8	39	Firmware Version	8	R	Identifies the version number of the device's EPROM level.
9	155	Key ID	11	C	Identifies the CA Key ID associated with the Signed Key being requested. This is a two byte ID followed by 9 spaces. This field is present only when a Signed PKI Key is being requested. Source: Device
<FS>					

[1]Required/Optional/Conditional

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

7. Impacted Data Elements

This section describes the existing data elements impacted by TransArmor processing:

- Account Number (Element No. 2)
- Card Discretionary Block Data (Element No. 12)
- Data Type Indicator (Element No. 24)
- Decline Code (Element No. 26)
- Firmware Version (Element No. 39)
- Hardware Version (Element No. 43)
- Load Type (Element No. 48)
- Message Format Version Identifier (Element No. 55)
- Number of Segments (Element No. 63)
- Pump/Lane Number (Element No. 79)
- Response Code (Element No. 83)
- Segment Length (Element No. 84)
- Segment Type (Element No. 85)
- Sequence Number (Element No. 86)
- Software Version (Element No. 96)
- Terminal Display/Printer Message (Element No. 101)
- Terminal Identifier (Element No. 102)
- Key ID (Element No. 155)
- Key Data Length (Element No. 156)
- Key Data (Element No. 157)

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Number:	2	**Name:**	Account Number
Character Type:	ANS	**Maximum Length:**	
		Non-TransArmor (all requests): 24 bytes	
		Non-TransArmor (all responses): 25 bytes	
		TransArmor	
		PKI Encryption and Tokenization:	
		Initial Request	425 bytes
		Initial Response	25 bytes
		Follow-on Request	50 bytes
		Follow-on Response	25 bytes
		Token Only:	
		Initial Request	425 bytes
		Initial Response	25 bytes
		Follow-on Request	425 bytes
		Follow-on Response	25 bytes
		TransArmor - VeriFone Edition	
		Encryption and Tokenization:	
		All Requests	425 bytes
		All Responses	25 bytes

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Representation:

Non-TransArmor—Initial Transaction—Request:

Format: 123456789012nn…nn

Where 123456789012nn…nn = a variable length (up to 24 alphanumeric bytes) Account Number.

Non-TransArmor—Initial Transaction—Response:

Format: 123456789012nnnnnnnnnnnnn

Where 123456789012nnnnnnnnnnnnn = a fixed length (25 alphanumeric bytes) Account Number.

Non-TransArmor—Follow-on Transaction—Request:

Format: 123456789012nn…nn

Where 123456789012nn…nn = a variable length (up to 24 alphanumeric bytes) Account Number.

Non-TransArmor—Follow-on Transaction—Response:

Format: 123456789012nnnnnnnnnnnnn

Where 123456789012nnnnnnnnnnnnn = a fixed length (25 alphanumeric bytes) Account Number.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Representation (*continued*):

TransArmor (PKI Encryption and Tokenization)—Initial Transaction—Request:

Format: TAPnnnnnnxyyyyyyyyyyyyzzzz[EDATA]

Where TAPnnnnnnxyyyyyyyyyyyyzzzz = the fixed length (25 alphanumeric bytes) TransArmor Header;

Followed by [EDATA] = the encrypted maximum variable length (up to 400 bytes) data block.

The content of the TransArmor Header is described as follows:

TAP = 3-byte TransArmor Indicator

Valid Value: TAP

nnnnnn = 6-byte Sequence Number of the transaction that matches the value in Element No. 86 (Sequence Number)

x = 1-byte [EDATA] Identifier that describes the content of [EDATA]

Valid Values: 1 = Encrypted Track1 data

2 = Encrypted Track2 data

3 = Encrypted manually-keyed PAN

yyyyyyyyyyy = 11-byte Key ID

zzzz = 4-byte Token Type ID

Valid Values: XXXX = Single use or MultiPay Token; 4-byte alphanumeric value assigned by First Data

The content of [EDATA] is described as follows:

[EDATA] = the encrypted maximum variable length (up to 400 bytes) data block.

TransArmor (PKI Encryption and Tokenization)—Initial Transaction—Response:

Format: 123456789012nn…nn

Where 123456789012nn…nn = the variable length (up to 25 alphanumeric bytes) retrieved Token

Note: The last four bytes of the token are the same as the PAN's last four bytes.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Representation (*continued*):

TransArmor (PKI Encryption and Tokenization)—Follow-on Transaction—Request:

Format: TAPnnnnnnxyyyyyyyyyyyzzzz[EDATA]

Where TAPnnnnnnxyyyyyyyyyyyzzzz = the fixed length (25 alphanumeric bytes) TransArmor Header;

Followed by [EDATA] = the maximum variable length (up to 25 bytes) Token retrieved in the original initial transaction Response.

The content of the TransArmor Header is described as follows:

TAP = 3-byte TransArmor Indicator

Valid Value: TAP

nnnnnn = 6-byte Sequence Number of the transaction that matches the value in Element No. 86 (Sequence Number)

x = 1-byte [EDATA] Identifier that describes the content of [EDATA]

Valid Values: 0 = Token

yyyyyyyyyyy = 11-byte Key ID

Note: There is no semicolon (;) in this value.

zzzz = 4-byte Token Type ID

Valid Values: XXXX = Single use or MultiPay Token; 4-byte alphanumeric value assigned by First Data

The content of [EDATA] is described as follows:

[EDATA] = the maximum variable length (up to 25 bytes) Token retrieved in the original initial transaction Response. The Token is included here.

TransArmor (PKI Encryption and Tokenization)—Follow-on Transaction—Response:

Format: 123456789012nn…nn

Where 123456789012nn…nn = the variable length (up to 25 alphanumeric bytes) retrieved Token

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Representation (*continued*):

TransArmor (Token Only)—Initial Transaction—Request:

Format: TAPnnnnnnxyyyyyyyyyyyyzzzz[EDATA]

Where TAPnnnnnnxyyyyyyyyyyyyzzzz = the fixed length (25 alphanumeric bytes) TransArmor Header;

The content of the TransArmor Header is described as follows:

TAP = 3-byte TransArmor Indicator

Valid Value: TAP

nnnnnn = 6-byte Sequence Number of the transaction that matches the value in Element No. 86 (Sequence Number)

x = 1-byte [EDATA] Identifier that describes the content of [EDATA]

Valid Values: 4 = Unencrypted data (Track1, Track2, manually-keyed PAN)

yyyyyyyyyyy = zero-filled (00000000000) (**Note**: A Key ID only applies when encrypted data is sent.)

zzzz = 4-byte Token Type ID

Valid Values: XXXX = Single use or MultiPay Token; 4-byte alphanumeric value assigned by First Data

TransArmor (Token Only)—Initial Transaction—Response:

Format: 123456789012nnnnnnnnnnnn

Where 123456789012nnnnnnnnnnnn = the fixed length (25 alphanumeric bytes) retrieved Token

Note: The last four bytes of the token are the same as the PAN's last 4 bytes.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Representation (*continued*):

TransArmor (Token Only)—Follow-on Transaction—Request:

Format: TAPnnnnnnxyyyyyyyyyyyyzzzz[EDATA]

Where TAPnnnnnnxyyyyyyyyyyyyzzzz = the fixed length (25 alphanumeric bytes) TransArmor Header;

The content of the TransArmor Header is described as follows:

TAP = 3-byte TransArmor Indicator

Valid Value: TAP

nnnnnn = 6-byte Sequence Number of the transaction that matches the value in Element No. 86 (Sequence Number)

x = 1-byte [EDATA] Identifier that describes the content of [EDATA]

Valid Values: 0 = Token

yyyyyyyyyyy = zero-filled (00000000000) (**Note**: A Key ID only applies when encrypted data is sent.)

zzzz = 4-byte Token Type ID

Valid Values: XXXX = Single use or MultiPay Token; 4-byte alphanumeric value assigned by First Data

TransArmor (Token Only)—Follow-on Transaction—Response:

Format: 123456789012nn…nn

Where 123456789012nn…nn = the variable length (up to 25 alphanumeric bytes) retrieved Token

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Representation (*continued*):

TransArmor - VeriFone Edition (Encryption and Tokenization)—Initial Transaction—Request:

Format: If card is swiped: TAPnnnnnnxyyyyy;yyyyyzzzz[EDATA]

or

If PAN is key-entered: TAPnnnnnnxyyyyy;yyyyyzzzz[EDATA]<fs>MMYY

Where TAPnnnnnnxyyyyy;yyyyyzzzz = the fixed length (25 bytes) TransArmor Header;

Followed by either:

[EDATA] = the TA-VE encrypted data block;

or

[EDATA]<fs>MMYY = If PAN is key-entered, the TA-VE encrypted data block; followed by a Field Separator, followed by the Expiration Date.

Content—TransArmor Header:

TAP = 3-byte TransArmor Indicator

Valid Value: TAP

nnnnnn = 6-byte Sequence Number of the transaction that matches the value in Element No. 86 (Sequence Number)

x = 1-byte [EDATA] Identifier that indicates the type of TA-VE support

Valid Value:
- 6 = TA-VE Encryption and Tokenization Support
- 7 = TransArmor VeriFone PIN pad will be uniquely identified with the pump/lane number for decryption purposes
- M = TA-VE with multiple terminal numbers and multiple logical lanes (TID comes from the BUYPASS ID, Terminal Number, and Lane)

yyyyy;yyyyy = 11-byte Key ID that includes a 5-byte Merchant Domain; followed by a semicolon (;); followed by a 5-byte Merchant Brand.

Note: Value will be provided by First Data during coding effort.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

zzzz = 4-byte Token Type ID

Valid Values: XXXX = Single use or MultiPay Token; 4-byte alphanumeric value assigned by First Data

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Content—[EDATA]:

Format: p…p:E…E

Note: This section applies to swiped transactions.

p…p = Processor Code

Variable-length field with a maximum length of 16 bytes, and a minimum length of 12 bytes. There is no padding.

Note: This is the merchant number used for settlement.

: = Colon

E…E = Encrypted Track Data (includes Expiration Date)

Note: Swiped transactions already contain Expiration Date within the track data, and BUYPASS counts the "=" or "^" sign as a Field Separator (<fs>).

Expiration Date will be encrypted. Do not alter the expiration date.

Content—[EDATA]<fs>MMYY:

Format: p…p:E…E<fs>MMYY

Note: This section applies to key-entered transactions.

p…p = Processor Code

Variable-length field with a maximum length of 16 bytes, and a minimum length of 12 bytes. There is no padding.

Note: This is the merchant number used for settlement.

: = Colon

E…E = Encrypted Keyed PAN (does not include Expiration Date). Maximum length of 400 bytes.

<fs> = Field Separator

Note: All keyed transactions and tokens must come to BUYPASS with a Field Separator between [EDATA] and Expiration Date.

MMYY = Expiration Date

Fixed length of 4 bytes

Default: 1249 (sent if no Expiration Date stored at the device)

Note: Expiration Date is sent to BUYPASS only when the Initial Transaction is a keyed transaction.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Representation (*continued*):

TransArmor - VeriFone Edition (Encryption and Tokenization)—Initial Transaction—Response:

Format: 123456789012nn…nn

Where 123456789012nn…nn = the variable length (up to 25 alphanumeric bytes) retrieved Token

Note: The last four bytes of the token are the same as the PAN's last four bytes.

Note: The unencrypted Expiration Date is returned in the Additional Data Segment (No. 112) in Field ID 013.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Representation (*continued*):

TransArmor - VeriFone Edition (Encryption and Tokenization)—Follow-on Transaction—Request:

Note: This is a TA-VE transaction submitted with a Token.

Format: TAPnnnnnnxyyyyy;yyyyyzzzz[EDATA]<fs>MMYY

Where TAPnnnnnnxyyyyy;yyyyyzzzz = the fixed length (25 bytes) TransArmor Header;

Followed by:

[EDATA]<fs>MMYY = the TA-VE token, followed by a Field Separator, followed by the clear Expiration Date.

Content—TransArmor Header:

TAP = 3-byte TransArmor Indicator

Valid Value: TAP

nnnnnn = 6-byte Sequence Number of the transaction that matches the value in Element No. 86 (Sequence Number)

x = 1-byte [EDATA] Identifier that indicates the type of TA-VE support

Valid Value: 0 = Token

yyyyy;yyyyy = zero-filled (00000;00000)

Note: A Key Id is not used.

zzzz = 4-byte Token Type ID

Valid Values: XXXX = Single use or MultiPay Token; 4-byte alphanumeric value assigned by First Data

Content—[EDATA]:

Format: E…E

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Content—[EDATA]:

Format:

Note: This section applies to Follow-on Transactions with token. The processor code is NOT sent in Follow-on transactions.

E…E = Token received in initial transaction or on file

<fs> = Field Separator

Note: All keyed and token transactions must come to BUYPASS with a Field Separator between (EDATA) and Expiration Date.

MMYY = Expiration Date

Fixed length of 4 bytes

Default: 1249 (sent if no Expiration Date stored at the device)

Note: Expiration Date is sent to BUYPASS only when the Initial Transaction is a keyed transaction.

The unencrypted Expiration Date is returned in Additional Data Segment (No. 112) in Field ID 013 in the original Transaction Response and must be present in Token Follow-on transactions.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Representation (*continued*):

TransArmor - VeriFone Edition (Encryption and Tokenization)—Follow-on Transaction—Response:

Note: This is a TA-VE transaction submitted with a Token.

Format: 123456789012nn…nn

Where 123456789012nn…nn = the variable length (up to 25 bytes) retrieved Token.

Note: The Expiration Date is returned in the Additional Data Segment (No. 112) in Field ID 013.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Purpose:

Non-TransArmor: Identifies the card or account.

TransArmor (Either Encryption Method): Identifies the 25-byte TransArmor header plus the 400-byte EDATA block that contains the encrypted/unencrypted information that identifies the account for TransArmor.

Note: In the case of the Initial Transaction Request for the PKI Encryption and Tokenization method, the first 8 bytes of EDATA must be composed of 2 zeros followed by the first 6 bytes of the Merchant Number assigned by BUYPASS. Please see the description for Element No. 102 (Terminal Identifier) for additional information about the Merchant Number.

Processing Rules:

For **non-TransArmor** and **TransArmor Token Only**, the POS device maintains the card's Account Number until it receives a response from the host.

For **TransArmor PKI Encryption and Tokenization**, the POS device must not maintain the card's Account Number after the initial card swipe.

For **Non-TransArmor and TransArmor**, if the Account Number is obtained from a Track 2 card reader, instead of being manually keyed, the Account Number and Card Discretionary Block Data should consist of all data read from the card between, but not including, the Start Sentinel and End Sentinel. Account Number is considered the data before the = character, and Card Discretionary Block Data is considered the data after the = character. The = character is used instead of a Field Separator.

For **Non-TransArmor and TransArmor**, if the Account Number is obtained from a Track 1 card reader (credit cards only), instead of being manually keyed, the Account Number and Card Discretionary Block Data should consist of all unaltered data read from the card between, but not including, the Start Sentinel and End Sentinel. Account Number is considered the data before the first ^ character, and Card Discretionary Block Data is considered the data after the first ^ character. The ^ character is used instead of a Field Separator. For **Non-TransArmor** – Account Number and Card Discretionary Block Data combined do not exceed 76 bytes for Track 1 or 37 bytes for Track 2.

For **TransArmor (Either Encryption Method)**, Account Number and Card Discretionary Block Data combined do not exceed 476 bytes for Track 1 or Track 2.

Note: Track data placed in the combined data elements of Account Number and Card Discretionary Block Data cannot exceed the total length of those data elements.

For **eParms enabled Terminals**, the terminal must include eParms data (variable length up to 300 characters). eParms includes information about the device as well as the status of the track/manually entered data. Once enabled, eParms data must be included with every transaction.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

eParms data must follow the format preserving data. Format is preserved encrypted data and eParms are separated by the field separator “|”.

Sample encrypted data from eParms enabled device is shown below:

3569994872982211=18122010231626951449|01010061250220150b284-109-
108160d028410910800084102663d9b97017e527861032419f22370e8126d809000000286d302097c81

Real Time Account Updater:

When a merchant receives Account Updater TransArmor Token (New TransArmor Token) in the original transaction response in the Real Time Account Updater Response Segment – Segment 155, the merchant can send either the new TransArmor token or the original TransArmor token in the ‘Follow-on’ transactions in this Account Number field as TAP request.

In the event of a ‘Retry’ of a ‘Follow-on’ transaction (Reversals, Completions, Cancellations, Void of Merchandise Return, and Timeout reversals) by the merchant, the merchant should use the same credentials for the retry as were used in the initial attempt.

Valid Codes/Values:

Credit, debit, and EBT cards: Please refer to the most recent version of *BUYPASS® Platform Issuer Identification Number (IIN) Reference Manual*.

Fleet cards: Please refer to the most recent version of *BUYPASS® Platform Petroleum Industry Processing Specifications*.

Stored Value cards: The length of the Account Number should be 16 or 19 bytes.

Note: For a First Data™ Premium Gift Card Transaction Response, the calculated card number is returned in the Account Number field.

Check Services: All MICR data (scanned or manually entered) up to 23 bytes are included in this field. This data must also be included in MICR Data (Element No. 122) of the Check Data Segment (Segment No. 110) .For details on the Check Data Segment, please refer to the latest version of ATL105 Message Format Specifications. The Account Number should contain the following keyed or swiped information:

Service Provider	Pos. Nos.	Data
SCAN (ETC)	1–23	MICR data or driver’s license
ECA/ TeleCheck® service	1–23	MICR data or driver’s license
Certegy (Formerly Telecredit)	1–24	Driver’s license

Loyalty cards: For a Loyalty Totals Request, Update Code value of "T", the words "TOTALS REQUEST" must appear in the Account Number

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

field. In any other loyalty advice transaction, the value is the actual Loyalty Account Number.

Speedpass transactions: For a Speedpass initiated request the Speedpass ID should appear in the Account Number area.

Speedpass ID	Length
DST	15 digits
Non-DST	24 digits

When manually keyed, the presentation of both the Account Number and the Sub-Account Number is required for Processing. The format of the Account Number Data Element is the complete account number followed by a single SPACE/BLANK character, followed by the 3 character Sub-Account Number.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Number:	12	**Name:**	Card Discretionary Block Data
Character Type:	AN	**Maximum Length:**	51 bytes

Representation:	Variable length of up to 51 alphanumeric characters
Purpose:	Identifies the card's expiration date and/or discretionary data.
Processing Rules:	The format varies, depending on the application. If the Account Number is obtained from a Track 2 card reader, instead of being manually keyed, the Account Number and Card Discretionary Block Data should consist of all data read from the card between, but not including, the Start Sentinel and End Sentinel. Account Number is considered the data before the = character, and Card Discretionary Block Data is considered the data after the = character. The = character is used instead of a Field Separator. If the Account Number is obtained from a Track 1 card reader (credit cards only), instead of being manually keyed, the Account Number and Card Discretionary Block Data should consist of all unaltered data read from the card between, but not including, the Start Sentinel and End Sentinel. Account Number is considered the data before the first ^ character, and Card Discretionary Block Data is considered the data after the first ^ character. The ^ character is used instead of a Field Separator. Account Number and Card Discretionary Block Data combined do not exceed 76 characters for Track 1 or 37 characters for Track 2. **Note:** Track data placed in the combined data elements of Account Number and Card Discretionary Block Data cannot exceed the total length of those data elements For all TransArmor token initiated transactions, the Expiration Date must be sent in clear in this field. **Real Time Account Updater:** When a merchant receives Account Updater Expiration Date (New Expiration Date) in the original transaction response in the Real Time Account Updater Response Segment – Segment 155, the merchant can send either the new expiration date or the original expiration date in the 'Follow-on' transactions in this Card Discretionary Block Data field. A merchant should send the expiration date corresponding to the original or the new TransArmor token that is being sent in the cardholder

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

	Account Number field – Data Element 2 in the Standard Message Data Segment 100. In the event of a 'Retry' of a 'Follow-on' transaction (Reversals, Completions, Cancellations, Void of Merchandise Return, and Timeout reversals) by the merchant, the merchant should use the same credentials for the retry as were used in the initial attempt.
Valid Codes/Values:	For a card whose number has been manually entered, the expiration date is in MMYY format, and the other required data depends upon the application. For a Certegy (Formerly Telecredit) transaction, the information consists of the 2-character alphabetical State Code (See Appendix D. Valid State Codes in the latest version of ATL105 Message Format Specifications) and the 6-character birth date (MMDDYY). If a driver's license number is used as identification for a SCAN (ETC) transaction, the information consists of the 2-character alphabetical State Code (See Appendix D. Valid State Codes in the latest version of ATL105 Message Format Specifications).

Number: 24 Name: Data Type Indicator

Character Type: AN Maximum Length: 1 byte

Representation:	Fixed length of one alphanumeric character	
Purpose:	Identifies specific characteristics of the following data in a host response.	
Processing Rules:	Required for TransArmor PKI Encryption and Tokenization.	
Valid Codes/Values:	**Code**	**Description**
	#	Identifies the Merchant Data Segment (No. DL1).
	!	Identifies the Dial String Data Segment (No. DL2).
	:	Identifies the Date and Time Data Segment (No. DL3).
	@	Identifies the Software Dial Load Data Segment (No. DL4).
	$	Identifies the Software IP Load Data Segment (No. DL5).
	\	Identifies the Store and Forward Data Segment (No. DL6).
	&	Indicates that a new Key and Key ID are being returned to a device using TransArmor PKI Encryption and Tokenization.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Number:	26	**Name:**	Decline Code
Character Type:	AN	**Maximum Length:**	2 bytes

Representation:	Fixed length of two alphanumeric characters
Purpose:	Identifies the reason for the decline.
Processing Rules:	BUYPASS defines this code. For some Decline Codes, a device can be programmed to display a decline message prompting intervention by a clerk/customer. For the other Decline Codes, a device displays a decline message and completes the transaction. For a TransArmor Token Only transaction that has been declined because BUYPASS cannot retrieve the necessary token for the associated PAN, please refer to the appendix indicated below for additional information.
Valid Codes/Values:	Any valid Decline Code. For a list of valid Decline Codes permitting clerk/customer intervention following a decline, please refer to Appendix H. Decline Codes Permitting Clerk/Customer Intervention in the latest version of ATL105 Message Format Specifications.

Number:	30	**Name:**	Download Indicator
Character Type:	N	**Maximum Length:**	1 byte

Representation:	Fixed length of one digit	
Purpose:	Indicates whether the device should request a load; and, if so, what type.	
Processing Rules:	BUYPASS leaves communications established so that a Load Request can be initiated. Required for TransArmor PKI Encryption and Tokenization. This value indicates the device must update its Key ID and Key because the current Key ID is due to expire or has expired.	
Valid Codes/Values:	**Code**	**Description**
	0	No Download Request.
	1	Indicates device should request a partial load.
	8	Indicates the device must update its Key ID and Key

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Number:	39	**Name:**	Firmware Version
Character Type:	AN	**Maximum Length:**	8 bytes

Representation:	Fixed length of eight alphanumeric characters
Purpose:	Identifies the device's firmware version at the customer location.
Processing Rules:	Required for Totals Requests, Table Load Request, and TransArmor PKI Encryption and Tokenization Load Request.
Valid Codes/Values:	

Number:	43	**Name:**	Hardware Version
Character Type:	AN	**Maximum Length:**	4 bytes

Representation:	Fixed length of four alphanumeric characters
Purpose:	Identifies the device's hardware version.
Processing Rules:	Required for Totals Requests, Table Load Request, and TransArmor PKI Encryption and Tokenization Load Request.
Valid Codes/Values:	Any valid hardware version number

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Number:	48	**Name:**	Load Type
Character Type:	A	**Maximum Length:**	1 byte

Representation: Fixed length of one alpha character

Purpose: Identifies the load type.

For TransArmor PKI Encryption and Tokenization, it indicates that Key information is requested in the Key and Key ID request.

Processing Rules: Required for TransArmor PKI Encryption and Tokenization.

Valid Codes/Values:

Codes	Description
P	Partial Load (Phone, Table)
D	Date and Time Load
K	TransArmor PKI Encryption and Tokenization Load Unsigned Key or Signed Key
S	Signing Key ID

Number:	55	**Name:**	Message Format Version Identifier
Character Type:	AN	**Maximum Length:**	6 bytes

Representation: Fixed length of six alphanumeric characters

Purpose: Identifies the type and release version of message format.

Processing Rules: All Financial Transaction Request messages, Totals Request messages, Loyalty Card Request messages, and Electronic Mail Request messages contain this data element in Field No. 1. It originates at the device.

For Financial Transaction Requests, the following data element and data segment always follow this data element:

- Element No. 63, Number of Segments
- Data Segment No. 100, Standard Message Data Segment

plus none, one, or more of the following data segments:

- Data Segment No. 101, Fleet Data Segment
- Data Segment No. 102, Product Code Data Segment
- Data Segment No. 103, EBT Data Segment
- Data Segment No. 104, Purchase Card Data Segment
- Data Segment No. 111, Variable Information Data Segment

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

- Data Segment No. 117, Speedpass Data Segment

For Financial Transaction Responses, the following data segment can follow the last data element in the message:

- Additional Information Data Segment (Data Segment No. 112)

For Totals Request messages, the following data element and data segment always follow this data element;

- Element No. 63, Number of Segments
- Data Segment No. 105, Totals Data Segment

For Loyalty Card Request messages, the following data element and data segment always follow this data element;

- Element No. 63, Number of Segments
- Data Segment No. 108, Loyalty Card Data Segment

For Electronic Mail Request messages, the following data element and data segment always follow this data element;

- Element No. 63, Number of Segments
- Data Segment No. 109, Electronic Mail Data Segment

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

For TransArmor PKI Encryption and Tokenization Load Request messages, the following data element and data segment always follow this data element

- Element No. 63, Number of Segments
- Data Segment No. 116, TransArmor Load Data Segment

Valid Codes/Values:

Value	Description
ATL105	Device/Controller (default)

Number: 63 Name: Number of Segments

Character Type: N Maximum Length: 2 bytes

Representation: Variable length of up to two digits

Purpose: Identifies the number of BUYPASS-defined data segments included in the Financial Transaction Request, Totals Request, Loyalty Card Transaction Request, and Electronic Mail Request.

Processing Rules: It originates at the device.

Financial Transaction Requests: This data element is always followed by Data Section No. 2 which contains Data Segment No. 100, Standard Message Data Segment, plus Data Section No. 3 which contains none, one, or more of the following data segments:

- Data Segment No. 101, Fleet Data Segment
- Data Segment No. 102, Product Code Data Segment
- Data Segment No. 103, EBT Data Segment
- Data Segment No. 104, Purchase Card Data Segment
- Data Segment No. 111, Variable Information Data Segment
- Data Segment No. 117, Speedpass Data Segment

Loyalty Card Transaction Requests: This data element is always followed by Data Section No. 2 which contains Data Segment No. 100, Speedpass Data Segment, plus Data Section No. 3 which contains Data Segment No. 108, Loyalty Card Data Segment, and Data Segment 114, SKU Data Segment.

ECA/ TeleCheck® Service Transaction Requests: This data element is always followed by Data Section No. 2 which contains Data Segment No. 100, Standard Message Data Segment, plus Data Section No. 3 which contains Data Segment No. 110, Check Data Segment, and Data Segment 111, Variable Information Data Segment.

Totals Requests: This data element is always followed by Data Segment No. 105, Totals Data Segment.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Electronic Mail Requests: This data element is always followed by Data Section No. 2, which contains Data Segment No. 109, Electronic Mail Data Segment.

Communications Test Requests: This data element is always followed by Data Section No. 2 which contains Element No. 120, Network Management Message.

TransArmor PKI Encryption and Tokenization: This data element is used for a Key and Key ID Load. In the requests, it is always followed by Data Section No. 2, which contains Data Segment No. 116, TransArmor Load Data Segment.

Valid Codes/Values:

Code	Description
1	Indicates that this data element is followed by only one of the following: • Data Segment No. 100 (Standard Message Data Segment) • Data Segment No. 105 (Totals Data Segment) • Data Segment No. 109 (Electronic Mail Data Segment) • Element No. 120 (Network Management Message) • Data Segment No. 116 (TransArmor® Load Data Segment) • Data Segment No. 118 (Proprietary Data Load Segment)
2	Indicates that this data element is followed by Data Segment No. 100 (Standard Message Data Segment) plus one of the following data segments: • Data Segment No. 101 (Fleet Data Segment) • Data Segment No. 102 (Product Code Data Segment) • Data Segment No. 103 (EBT Data Segment) • Data Segment No. 104 (Purchase Card Data Segment) • Data Segment No. 111 (Variable Information Data Segment) • Data Segment No. 117, (Speedpass Data Segment)
3	Indicates that this data element is followed by Data Segment No. 100 (Standard Message Data Segment); plus two of the following data segments: • Data Segment No. 101 (Fleet Data Segment) • Data Segment No. 102 (Product Code Data Segment) • Data Segment No. 103 (EBT Data Segment) • Data Segment No. 104 (Purchase Card Data Segment) • Data Segment No. 108 (Loyalty Card Data Segment) • Data Segment No. 111 (Variable Information Data Segment) • Data Segment No. 114 (SKU Data Segment)

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

- Data Segment No. 117, (Speedpass Data Segment)

4 Indicates that this data element is followed by Data Segment No. 100 (Standard Message Data Segment); plus three of the following data segments:

- Data Segment No. 101 (Fleet Data Segment)
- Data Segment No. 102 (Product Code Data Segment)
- Data Segment No. 103 (EBT Data Segment)
- Data Segment No. 104 (Purchase Card Data Segment)
- Data Segment No. 110 (Check Data Segment)
- Data Segment No. 111 (Variable Information Data Segment)
- Data Segment No. 113 (ECA/ TeleCheck® Data Segment)
- Data Segment No. 117, (Speedpass Data Segment)

5 Indicates that this data element is followed by Data Segment No. 100 (Standard Message Data Segment); plus four of the following data segments:

- Data Segment No. 101 (Fleet Data Segment)
- Data Segment No. 102 (Product Code Data Segment)
- Data Segment No. 103 (EBT Data Segment)
- Data Segment No. 104 (Purchase Card Data Segment)
- Data Segment No. 111 (Variable Information Data Segment)

6 Indicates that this data element is followed by Data Segment No. 100 (Standard Message Data Segment); plus all of the following data segments:

- Data Segment No. 101 (Fleet Data Segment)
- Data Segment No. 102 (Product Code Data Segment)
- Data Segment No. 103 (EBT Data Segment)
- Data Segment No. 104 (Purchase Card Data Segment)
- Data Segment No. 111 (Variable Information Data Segment)

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Number:	79	**Name:**	Pump/Lane Number
Character Type:	N	**Maximum Length:**	2 bytes

Representation:	Variable length of up to two digits
Purpose:	Identifies the pump or lane number at the point-of-sale location.
Processing Rules:	Required on all fuel transactions. If Data Element 2 (Account Number) EDATA Identifier is of value 7, then this data element will be used as pump/lane number for decryption purposes only.
Valid Codes/Values:	1–99

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Number:	83	**Name:**	Response Code
Character Type:	AN	**Maximum Length:**	1 byte

Representation: Fixed length of one alphanumeric character

Purpose: For financial transactions, it identifies an approval with purchase/capture, an approval without purchase/capture, or a decline.

For totals, it identifies an approval, an approval with electronic mail retrieval pending, decline with electronic mail retrieval pending, approved with electronic mail retrieval pending and proprietary data pending, declined with electronic mail retrieval pending and proprietary data pending, approved with proprietary data pending, or declined with proprietary data pending.

For electronic mail, it identifies an approval or an approval with more mail pending.

For communications test, it identifies an approval.

For a TransArmor PKI Encryption and Tokenization transaction, it identifies one of the following:

- The transaction is an approved TransArmor PKI Encryption and Tokenization transaction.
- The transaction is a rejected TransArmor PKI Encryption and Tokenization transaction.
- The merchant is not a TransArmor merchant.

Processing Rules:

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Valid Codes/Values:	Transaction Code	Description
	0	Approved—Purchase/Capture
	1	Approved—Communications Test
		Declined—All other transactions
	2	Approved—Authorization only
	3	Approved—Authorization only with AVS
	4	Approved—Purchase/Capture with AVS
	5	Approved—Totals
	6	Approved—Totals with electronic mail retrieval pending
	7	Declined—Totals with electronic mail retrieval pending
	8	Approved—Electronic mail
	9	Approved—Electronic mail and more mail pending
	B	Approved—Totals with electronic mail retrieval pending and proprietary data pending
	C	Declined—Totals with electronic mail retrieval pending and proprietary data pending
	D	Approved—Totals with proprietary data pending
	E	Declined—Totals with proprietary data pending
	F	Approved—Partial approval
	G	Approved—Electronic mail and end of proprietary data block
	H	Approved—Proprietary data retrieval, more pending
	J	Approved—Electronic Mail, reset
	K	Approved—TransArmor Load
	L	Rejected for one of the following reasons— 1) Merchant is not TransArmor enabled 2) Signing Key is invalid or outdated
	M	Declined—Totals with Proprietary Host Discount data pending
	N	Approved—Totals with Proprietary Host Discount data pending

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

O	Approved—Proprietary data load, more pending
P	Successful Signing/Signed Key Load Request
S	Declined—Retry the transaction.
T	Approved—Proprietary data load, no more data pending
U	Declined—Proprietary data load, no more data pending
V	Declined—Totals with Proprietary Custom Receipt Text data pending
W	Approved—Totals with Proprietary Custom Receipt Text pending
X	Declined—Proprietary data load, proceed to next pending Prompt Code
Y	Approved—Proprietary data load, proceed to next pending Prompt Code

Note: In some instances, proprietary data can be downloaded.

If a TransArmor Key Load (Signing/Signed/Unsigned) request is declined, the last element in the message will be Data Element 83 (Response Code).

Number: 84 Name: Segment Length

Character Type: N Maximum Length: 4 bytes

Representation: Variable length of three or four digits

Note: Segment Length has a length of four digits in only one instance—when it is used in the transmission of the SKU Data Segment (No. 114).

Purpose: Identifies the length of the segment in which it appears.

Processing Rules: This figure is calculated using the lengths of all data elements in the segment plus all Field Separators/Product Data Field Delimiters related to those elements. (Please refer to chapter 15, “Data Segment Formats,” in the latest version of ATL105 Message Format Specifications for additional information about specific data segments—including their composition, data element lengths, Field Separators, and Product Data Field Delimiters.)

**This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.**

Note: Segment Length can only have a length of 4 digits when transmitting the SKU Data Segment (No. 114). It cannot have a length of 4 digits when sending any other segment.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Valid Codes/Values:	Data Length	Data Segment
	001–218	Standard Message Data Segment (No. 100)
	001–61	Fleet Data Segment (No. 101)
	001–381	Product Code Data Segment (No. 102)
	001–3334	EBT Data Segment (No. 103)
	001–86	Purchase Card Data Segment (No. 104)
	001–354	Totals Data Segment (No. 105)
	001–142	Loyalty Card Data Segment (No. 108)
	001–232	Electronic Mail Data Segment (No. 109)
	001–168	Check Data Segment (No. 110)
	001–999	Variable Information Data Segment (No. 111)
	001–999	Additional Information Data Segment (No. 112)
	001–156	ECA/ TeleCheck® Data Segment (No. 113)
	0001–1010	SKU Data Segment (No. 114)
	01–50	TransArmor Load Data Segment (No. 116)
	01-53	Speedpass Data Segment (No. 117)
	0001-3800	Proprietary Data Load Segment (No. 118)
	001-493	Totals with Proprietary Data Load Data Segment (No. 119)

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Number:	85	**Name:**	Segment Type
Character Type:	N	**Maximum Length:**	3 bytes

Representation: Fixed length of three digits

Purpose: Identifies the type of segment being formatted in a Financial Transaction request, a Totals request, a Loyalty Card transaction request, Electronic Mail request, or a TransArmor PKI Encryption and Tokenization Load Request.

For TransArmor PKI Encryption and Tokenization, this value indicates that the segment contains Key and Key ID information.

Processing Rules: Required for the TransArmor PKI Encryption and Tokenization Load Request.

Valid Codes/Values:

Code	Description
100	Data Segment No. 100, Standard Message Data Segment
101	Data Segment No. 101, Fleet Data Segment
102	Data Segment No. 102, Product Code Data Segment
103	Data Segment No. 103, EBT Data Segment
104	Data Segment No. 104, Purchase Card Data Segment
105	Data Segment No. 105, Totals Data Segment
106	Data Segment No. 106 is reserved for proprietary use.
107	Data Segment No. 107 is reserved for proprietary use.
108	Data Segment No. 108, Loyalty Card Data Segment
109	Data Segment No. 109, Electronic Mail Data Segment
111	Data Segment No. 111, Variable Information Data Segment
112	Data Segment No. 112, Additional Information Data Segment
116	Data Segment No. 116, TransArmor® Load Data Segment
117	Data Segment No 117, Speedpass Data Segment
118	Data Segment No 118, Proprietary Data Load Segment
119	Data Segment No. 119, Totals with Proprietary Data Load Load Data Segment

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Number:	86	**Name:**	Sequence Number
Character Type:	N	**Maximum Length:**	6 bytes

Representation:	Fixed length of six digits
Purpose:	**Note:** Identifies a transaction for the life of the transaction. Sequence Number must persist in all the subsequent transactions that are associated with the initial authorization request. In the case of a preauthorized transaction, the Sequence Number in the Purchase/Capture identifies the Sequence Number in the transaction's Authorization Only request.
Processing Rules:	Each transaction has a unique Sequence Number. It is included on all transaction types, and originates at the device. In the case of a preauthorized transaction, the Sequence Number in the Purchase/Capture is the same Sequence Number as in the transaction's Authorization Only request. In the case of a purchase reversal/void or time-out reversal transaction, the Sequence Number must be the same as in the original transaction request. In the case of an Authorization Only Reversal, this element—present in the original transaction—must be present and identical in the Authorization Only Reversal transaction. (**Note**: The Authorization Only Reversal transaction enables a merchant to reverse the following credit card transactions within required timeframes: POS Authorization Only [Transaction Type code "3"]); POS Mail/Phone Authorization Only [Transaction Type code "B"]); and CAT Authorization Only [Transaction Type code "5"]). For TransArmor PKI Encryption and Tokenization, this element is included in a new data segment (No. 116 [TransArmor® Load Data Segment]).
Valid Codes/Values:	000001–999999

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Number:	96	**Name:**	Software Version
Character Type:	AN	**Maximum Length:**	8 bytes

Representation:	Fixed length of eight alphanumeric characters
Purpose:	Identifies the device's software version at the customer location.
Processing Rules:	Required for Totals Request, Table Load Request, and TransArmor PKI Encryption and Tokenization Load Request.
Valid Codes/Values:	

Number:	101	**Name:**	Terminal Display /Printer Message
Character Type:	AN	**Maximum Length:**	31 bytes

Representation:	Fixed length of 31 alphanumeric characters; space-filled
Purpose:	Identifies a device display or a printer message for the merchant from BUYPASS. **Examples.** *Fill_out_change_of_address_form* *Get_driver's_license_number.___* *Approved_990001_* *PASSWORD_ERR_5A_* In the first example, there are 26 characters and 5 spaces (indicated by the _ character) for a total of 31 alphanumeric characters. In the second example, there are 25 characters and 6 spaces (indicated by the _ character) for a total of 31 alphanumeric characters. In the third example, there are 14 characters and 2 spaces (indicated by the _ character) for a total of 16 alphanumeric characters. In the fourth example, there are 13 characters and 3 spaces (indicated by the _ character) for a total of 16 alphanumeric characters.
Processing Rules:	Print what is sent. This data element can be repeated up to a maximum of three times in a Financial Transaction Response. The number of times it is repeated is indicated by the Number of Print Lines value. This data element cannot be repeated in a Declined Totals Response.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Required for both the TransArmor PKI Encryption and Tokenization and TransArmor Token Only Financial Transaction Response in which a Token has not been retrieved from the host.

Valid Codes/Values: Thirty-one alphanumeric characters; space-filled

For TransArmor, the message indicates that BUYPASS has not been able to retrieve a token for the associated PAN for a TransArmor transaction.

Number: 102 Name: Terminal Identifier

Character Type: AN Maximum Length: 22 bytes

Representation: Variable length up to 22 alphanumeric characters

Purpose: Identifies the transaction device by Device Type, State Code (ANSI® Code), BUYPASS Merchant Number, and Device Number.

Processing Rules: Required for all transactions and messages.

The Terminal Identifier must be assigned by and set up at BUYPASS.

Note: In the case of a Table Load Request, Phone Load Request, Date and Time Load Request, or Software Load Request, the field length is 13 bytes; otherwise, the field length is a maximum of 22 bytes.

For the TransArmor PKI Encryption and Tokenization Load Request, the first two characters (Device Type) of this element's value must be "++" regardless of the actual device type.

Valid Codes/Values:

Device Type (Pos. Nos. 1–2)

Any valid 2-character Device Type assigned by BUYPASS personnel.

For the TransArmor PKI Encryption and Tokenization Load Request, this element's value must be "++" regardless of the actual device type.

State Code (Pos. Nos. 3–4)

Any valid 2-character State Code (ANSI® Code). Please see Appendix D. Valid State Codes, in the latest version of ATL105 Message Format Specifications for a list of valid ANSI® Code entries.

BUYPASS Merchant Number (Starts immediately in Pos. No. 5.)

Any valid 6- to 15-character BUYPASS Merchant Number assigned by BUYPASS personnel. The Device Number immediately follows the BUYPASS Merchant Number.

Note: If using TransArmor, the first 6 digits of the Merchant Number are used in EDATA. Please refer to the description of Element No. 2 (Account Number) for additional information.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Device Number (Starts immediately after the BUYPASS Merchant Number.)

Any valid 3-character Device Number assigned by BUYPASS personnel.

Number:	155	**Name:**	Key ID
Character Type:	AN	**Maximum Length:**	11 bytes

Representation:	Fixed length of 11 alphanumeric bytes
Purpose:	Used in TransArmor PKI Encryption and Tokenization to identify the Key ID associated with the encryption Key included in Element No. 157 (Key Data) in the TransArmor Load Response.
Processing Rules:	Required in the TransArmor Load Response. Once requested and approved, this Key ID must be used for all subsequent TransArmor transactions from this device.
Valid Codes/Values:	Any valid Key ID (including CA Key ID that is returned in response to a Signing Key request).

Number:	156	**Name:**	Key Data Length
Character Type:	N	**Maximum Length:**	3 bytes

Representation:	Maximum variable length of 3 bytes
Purpose:	Used in TransArmor PKI Encryption and Tokenization to identify the length of the following field (Element No. 157 [Key Data]) in the TransArmor Load Response.
Processing Rules:	Required in the TransArmor Load Response.
Valid Codes/Values:	000–999

Number:	157	**Name:**	Key Data

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Character Type: AN Maximum Length: 999 bytes

Representation:	Variable maximum length of 999 alphanumeric characters
Purpose:	Identifies either the new Encryption Key and the Key ID (Element No. 155) associated with it in the TransArmor Load Response or the Error Message.
Processing Rules:	Required in the TransArmor Load Response. For an approved TransArmor transaction, the maximum length of Signing and Signed Keys can be 999 bytes. For a declined TransArmor transaction, a message is returned in this field and displays at the device.
Valid Codes/Values:	Any valid key.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

8. Appendix A. TransArmor VeriFone Edition: Responses to Administrative Command

The following table contains a list of TransArmor - VeriFone Edition (Encryption and Tokenization) — BUYPASS Front End ATL105 Responses to Administrative Commands:

ATL105 Responses to TransArmor Administrative Commands		
RegiStart Responses from BUYPASS Front End to VF POS:		
Response Type	**Decline Code**	**Terminal Display/Printer Message**
Successful RegiStart	Spaces	“VERISHIELD ACTIVATION SUCCESS”
Failed RegiStart	4M	“PLEASE RETRY 4M"
Advance DDK Responses from BUYPASS Front End to VF POS:		
Successful Advance DDK	Spaces	“VERISHIELD ACTIVATION SUCCESS”
Failed Advance DDK	4M	“PLEASE RETRY 4M"
Update Settings Responses from BUYPASS Front End to VF POS:		
Successful Update Settings	Spaces	“VERISHIELD ACTIVATION SUCCESS”
Failed Update Settings	4M	“PLEASE RETRY 4M"

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.